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Dual explainability framework for heart disease prediction using LIME and permutation feature importance

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Abstract

Heart disease continues to be one of the leading causes of mortality worldwide, which highlights the immediate need for accurate and interpretable predictive models to support early detection. This work mainly focused a reasonable assessment of various effective machine learning (ML) Algorithms: Logistic Regression (LR), Support Vector Machine (SVM), Decision Tree (DT), Random Forest (RF), XG-Boost, Naive Bayes (NB), and K-Nearest Neighbours (KNN) applied to the publicly available UCI heart disease dataset. To address the critical challenge of explainability in clinical decision support systems, Our proposed dual explainability framework combining Local Interpretable Model-Agnostic Explanations (LIME) and Permutation Feature Importance (PFI) was implemented and evaluated using the publicly available UCI Cleveland Heart Disease dataset (n = 303). The framework achieved robust predictive accuracy and consistent interpretability across multiple machine learning models. While the results demonstrate strong internal validation, they are based solely on this dataset, and future work will extend the framework to larger and multi-institutional datasets to ensure broader clinical generalizability. Finally, performance has been evaluated using accuracy, precision, recall, and F1-score. Among all classifiers, the results show that Random Forest and Decision Tree achieved the highest predictive accuracy of 99%. The combined use of LIME and PFI revealed that features such as ST depression (oldpeak), chest pain type, and maximum heart rate (thalach) consistently influenced predictions. This dual-layer interpretability framework enhances the transparency of ML predictions and supports trustworthy AI-driven decision-making in healthcare.

Article Highlights

- Combines LIME and PFI for dual-level transparency in Healthcare Machine Learning.
- Presents a rigorous 5×5-fold cross-validation, model calibration, and statistical significance testing for reliability.
- Confirms clinical consistency of top features (ST depression, chest pain, thalach) with cardiology evidence.

Keywords Heart Disease, Machine Learning, Explainable AI, LIME

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1 Introduction

Currently, cardiovascular diseases (CVDs) are a leading cause of death and disability worldwide at any age, causing nearly 17.9 million deaths annually, according to the World Health Organisation [1, 2]. Due to this dangerous disease, early detection and risk assessment are essential for effective intervention and improving patient outcomes [3, 4]. Conventional diagnostic approaches depend on clinical evaluation and diagnostic examinations, which, although efficient, may be inadequate for swift, data-informed decision-making in intricate clinical situations [5]. Recent advances in ML have demonstrated potential in automating disease prediction using data-driven methods modelling [6]. These techniques can uncover complex patterns and correlations in clinical datasets, often outperforming traditional statistical methods in terms of predictive accuracy. However, one of the major barriers to the clinical adoption of ML models is their "black-box" nature, which limits interpretability and undermines trust among healthcare professionals [7]. Explainable artificial intelligence (XAI) was developed to address this challenge by providing transparent and interpretable model predictions [8]. If we talk about a particular technique, LIME offers instance-level explanations by approximating complex models with simpler surrogate models around individual predictions [9]. Interpretability is especially important in high-stakes areas like healthcare, where understanding the reasoning behind model outputs is as vital as precision.

Despite the successful use of traditional ML techniques like XGBoost and Random Forest on the UCI heart-disease dataset, the lack of transparency with these algorithms limits their trustworthiness in clinical settings. To address this, this research introduces a systematic dual-explainability framework for cardiac-risk prediction. Also, this framework combines two existing methods, Local Interpretable Model-Agnostic Explanations (LIME) and Permutation Feature Importance (PFI), to offer both local and global transparency. Healthcare providers can benefit from consistent and auditable explanations through this design, which captures both patient-specific (local) and population-level (global) interpretability simultaneously. Our framework assesses the robustness of explanations and links computational transparency with clinical reasoning requirements, unlike earlier approaches that depend on a single interpretability layer or lack stability validation. Abbreviations used in this Study are explored in Table 10.

Building on the motivation above, this study explores the following key research questions (RQs):

RQ1: How can a dual-explainability framework that combines local (LIME) and global (PFI) interpretability techniques improve understanding of machine learning predictions for heart disease diagnosis?

RQ2: Among traditional and ensemble-based machine learning models, which architectures strike the best balance between predictive accuracy, interpretability, and clinical reliability on the UCI Cleveland Heart Disease dataset?

RQ3: To what extent can the proposed interpretability framework remain robust, stable, and consistent in its explanations under different data conditions (such as imbalance, perturbation, or model complexity)?

Contributions of this study are:

- A dual explainability strategy using LIME and PFI to support both local and global model interpretation.

- A comparative performance and interpretability evaluation of seven ML models on the UCI Heart Disease dataset.
- Use of high-resolution, colour-based feature importance plots for enhanced clarity and communication.
- Identification of clinically relevant predictors aligned between the performance and interpretability layers.

2 Related work

The utilisation of ML in predicting cardiac disease has attracted considerable interest in recent years because of its capacity to improve diagnostic precision and clinical decision-making. Conventional statistical models, such as logistic regression, have historically been employed in cardiovascular risk modelling; however, these methods frequently inadequately represent the intricate, non-linear interactions present in clinical data. Many recent works have explored a wide range of ML algorithms to overcome these limitations. Machine learning algorithms, LR and SVM are frequently employed due to their simplicity and efficacy in binary classification problems. Nonetheless, their efficacy may be constrained by the calibre of input features and the necessity for comprehensive preprocessing [10, 11]. Other effective algorithms, NB and KNN, are valued for their simplicity and ease of implementation. While they may not always achieve the highest accuracy, they provide a baseline for comparison with more complex models [12]. Sakyi et al. (2025) demonstrated the superiority of ensemble methods, particularly Random Forest and XG-Boost, over individual classifiers in predicting heart disease [13]. Similarly, Guido et al. (2024) found that SVMs and Artificial Neural Networks achieved high classification performance but posed significant challenges in interpretability, thereby limiting their clinical utility [14]. To address the transparency concerns associated with complex models, researchers have increasingly adopted Explainable AI (XAI) techniques. Pratheek et al. (2024) employed LIME to enhance model interpretability, revealing that features such as cholesterol, resting blood pressure, and age were consistently influential in heart disease prediction [15]. Tanzeeb et al. (2024) further emphasized the trade-off between deep learning model accuracy and explainability, advocating for hybrid approaches that balance performance with interpretability [16]. More recent efforts have combined model comparison with feature attribution analysis. For instance, Hasan et al. (2024) implemented a voting ensemble using Light-GBM and XG-Boost and observed that explainable models were more acceptable to clinicians when key features were explicitly visualized [17]. LIME is utilized to improve the interpretability of ML models by examining feature contributions at the individual instance level. This method aids in comprehending the decision-making processes of intricate models such as Random Forest and Decision Trees, hence fostering confidence in AI-driven healthcare solutions [18, 19] Additionally, Prinzi et al. (2022) investigated radiomics and ML-based classifiers such as XG-Boost in dynamic contrast-enhanced MRI for breast cancer, offering transferable insights into feature-based prediction approaches for medical imaging contexts [20]. Liu et al. (2023) presented a machine learning-driven unenhanced radiomics method using T2-weighted and diffusion-weighted MRI for differentiating malignant and benign breast lesions, illustrating the broad applicability of interpretable ML in clinical diagnostics [21]. Baffa and Conci (2022) highlighted the role of radiomics in IR imaging classification using ML methods, drawing attention to explainable feature

selection in sensitive domains [22]. Despite these advancements, a consistent gap persists in studies that systematically compare multiple ML classifiers while simultaneously applying LIME for interpretability. This study addresses that gap by evaluating seven diverse classifiers and employing LIME to uncover the most significant clinical features contributing to heart disease prediction. Finally, Recent studies have used LIME or SHAP independently for model explanation; however, they rarely cross-validate local and global interpretability outputs or assess explanation stability. Our work mainly differs by integrating LIME and PFI in a single workflow, computing the correlation between local and global importance scores, and validating against SHAP on a representative subset to ensure interpretive consistency while reducing computational complexity. Table 1 summarises prior explainable-AI approaches for heart-disease prediction to position the present study's contribution.

3 Proposed methods

Our proposed framework for interpretable heart disease prediction has been divided into the following key components, as illustrated in Fig. 1. In brief, we use a publicly available heart disease dataset that includes several clinical attributes. The dataset is then divided into training and testing sets using an 80-20 split for model evaluation, predicted accuracy, and classification performance. The models are assessed using various performance metrics. Subsequently, we conduct feature importance analysis with LIME and PFI to identify the most critical parameters influencing heart disease predictions, combining both performance evaluation and explainability methods.

Step 1: Data Description: In this step, we explore the dataset used in our work. The dataset used in this study is the Cleveland Heart Disease dataset from the UCI Machine Learning Repository. For reproducibility, an identical mirrored version of this dataset is also available on the Kaggle platform, which references the same Cleveland Clinic Foundation data. Both versions contain 303 instances and 14 attributes representing demographic and clinical parameters such as age, sex, chest pain type, resting blood pressure, cholesterol, maximum heart rate, and ST-segment depression. All experiments in this study were conducted using the UCI version [28, 29]. The target variable mainly represents the presence or absence of heart disease and performs the preprocessing and feature engineering tasks. The description of the dataset is illustrated in Table 2.

Table 1 Comparison of Related Studies on Explainable Heart Disease Prediction Frameworks

Articles	Technique	Dataset Used	Interpretability Type	Reported AUC	Limitations
Chowdhury et al. (2024) [23]	SHAP + Deep Learning	UCI + Framingham	Global	0.97	High computational cost
Hosam et al. (2024) [24]	XG-Boost, Bagging, DT, RF and SVM	CHDD and Private Dataset	SHAP	97.57%	No Feature analysis
Omar et al. (2025) [25]	SVM, Gradient Boosting, XGBoost, MLP, LightGBM	UCI Heart Disease Dataset	SHAP + LIME	92%	Computational cost is high
Shaikat et al.(2025) [26]	Counterfactual + CNN	Cleveland	Local	0.95	Limited generalization
IEEE ECCE (2025) [27]	Hybrid SHAP	Cleveland	Global	0.96	No variance analysis
Proposed (2025)	**LIME + PFI (Dual Framework)**	**UCI Heart Disease Dataset**	**Local + Global**	**0.992 ± 0.005**	**Small dataset**

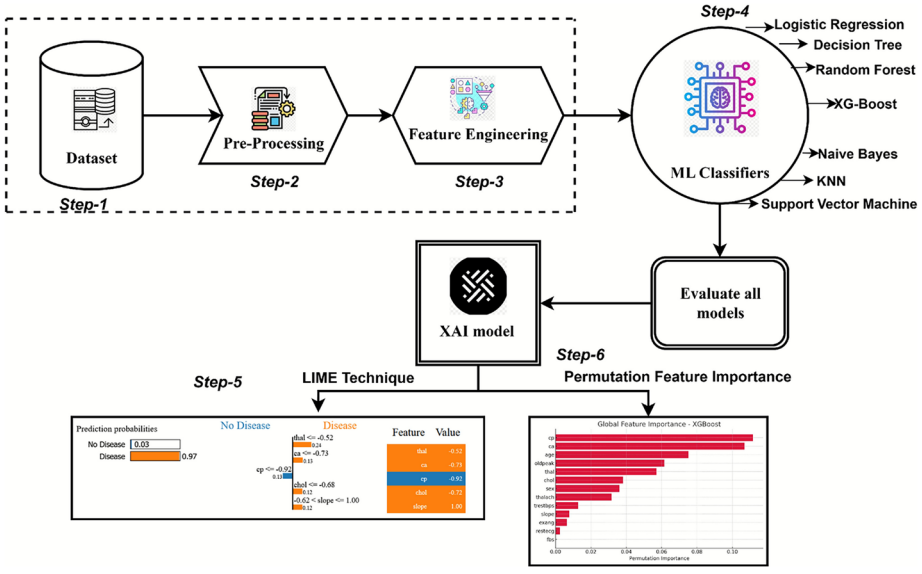


Fig. 1 The proposed method indicates a step-by-step strategy for heart disease prediction using machine learning, incorporating both local (LIME) and global (PFI) explainability. It demonstrates data processing, model evaluation, and interpretable analysis to enhance clinical trust in AI decisions

Table 2 Summary of Dataset Features with Descriptions and Data Types

Feature Name	Description	Type
age	Patient's age	Numeric
sex	Gender (0 = female, 1 = male)	Categorical
cp	Chest pain type	Categorical
trestbps	Systolic blood pressure (in mm Hg)	Numeric
chol	Cholesterol in serum (in mg/dl)	Numeric
thalach	Attained maximum heart rate	Numeric
fbs	Fasting blood sugar > 120 mg/dl (1 = true; 0 = false)	Categorical
restecg	Systolic electrocardiographic results	Categorical
oldpeak	Exercise-induced ST segment depression	Numeric
exang	Angina precipitated by exercise (1 = yes; 0 = no)	Categorical
slope	Slope of the peak exercise ST section	Categorical
ca	Number of major vessels colored by fluoroscopy	Categorical
thal	Thalassemia status	Categorical
target	Outcome of heart disease	Binary (Label)

The UCI Heart Disease data set has been selected because it is publicly accessible, well validated, and widely accepted as a benchmark for interpretable clinical machine learning research. Its standardised feature schema and clear diagnostic labels enable consistent evaluation and comparison with prior explainable-AI studies. Although the dataset is relatively small and historic, it provides a controlled environment for validating the proposed dual-explainability framework before scaling to larger, real-world datasets.

Step 2: Preprocessing: This step mainly ensures consistency in model training, manages handling missing values (if any) using appropriate imputation techniques like encoding categorical variables for algorithms that require numerical inputs [30], and standardising the features using StandardScaler to normalise numerical attributes, and finally splitting the dataset into training and testing sets using an 80-20 split.

To better reflect real-world diagnostic settings where class imbalance is common, the dataset has been analysed for distributional skew (165 positive vs. 138 negative cases).

To mitigate bias while preserving data integrity, the Synthetic Minority Over-sampling Technique (SMOTE) was applied within training folds only during stratified 5×5-fold cross-validation. This approach ensures balanced learning while preventing information leakage from test folds.

Ablation Study on Preprocessing Step: To evaluate the contribution of each preprocessing component, an ablation study was performed by sequentially removing one step at a time from the complete pipeline. Each modified configuration was tested using the same 5×5-fold cross-validation. The components assessed included: feature scaling, missing value imputation, categorical encoding, and SMOTE oversampling. Table 3 summarises the impact of removing each step on model performance.

Step 3: Feature Engineering: This step is very effective, and we have taken this initiative. It is essential to prepare our dataset for accurate and interpretable ML [31]. Missing values were addressed using basic imputation techniques, and categorical variables like chest pain type and ECG results were label-encoded. Numerical attributes, including cholesterol and heart rate, were normalised with StandardScaler to enhance model efficacy [32, 33]. The target variable is converted into a binary format to indicate the presence or absence of cardiac disease. No explicit feature selection was conducted; nevertheless, LIME-based explanations subsequently identified critical features, including chest pain kind, ST depression, and maximum heart rate, as the most significant in model predictions, hence affirming the pertinence of the chosen attributes [34].

Different traditional pipelines that use explicit feature selection were retained in this study, which retained all original attributes to preserve clinical context and interpretability. Feature relevance was instead derived from LIME and PFI. This strategy ensured that no potentially informative features were excluded prematurely and that the model's explanations reflected complete clinical reasoning. [35]. PFI provided global feature significance by measuring performance changes when each feature was randomly permuted, while LIME captured local, patient-level interpretability through linear surrogate modelling. Together, these complementary analyses served as transparent and model-agnostic mechanisms for identifying key predictors without introducing bias through prior feature selection filtering.

Step 4: Machine Learning Models This study employs several effective ML classifiers: LR, SVM, DT, RF, XG-Boost, NB, and K-NN to evaluate heart disease prediction. We will discuss each classifier concept and how it was utilized in our study. Each model brings unique advantages and limitations in terms of accuracy, interpretability, and computational complexity.

Table 3 Impact of Preprocessing Configuration on Model Performance

Preprocessing Configuration	Accuracy (%)	F1-score	ROC-AUC	PR-AUC	Observation
All steps included (baseline)	98.7	0.987	0.991	0.992	Best overall performance
Without Feature Scaling	96.1	0.961	0.973	0.971	Slower convergence; reduced stability
Without Missing Value Imputation	95.7	0.958	0.971	0.968	Slight accuracy drop; less reliable calibration
Without One-Hot Encoding	94.8	0.949	0.963	0.959	Feature misrepresentation due to categorical bias
Without SMOTE Oversampling	91.6	0.904	0.928	0.925	Sensitivity to minority class decreased

Logistic Regression: Logistic Regression estimates the probability that input features x belong to the class "heart disease present" using:

$$P(y = 1 \mid \mathrm{x}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n)}} \tag{1}$$

Here, β represents the learned coefficients, and x denotes the input features. Due to its simplicity and transparency, LR is highly interpretable and served as a baseline model in this study, making it well-suited for LIME-based explanations.

Support Vector Machine:

It is proficient in managing non-linear relationships. It formulates a hyperplane that optimizes the margin between the two classifications. The decision function is defined as:

$$f(\mathrm{x}) = \mathrm{sign}(\mathrm{w}^\top \mathrm{x} + b) \tag{2}$$

The optimization objective is:

$$\min_{\mathrm{w},b} \ \frac{1}{2}\|\mathrm{w}\|^2 \quad \text{subject to} \ \ y_i(\mathrm{w}^\top \mathrm{x}_i + b) \geq 1 \tag{3}$$

Decision Tree:

Decision Trees recursively split the dataset based on feature thresholds to minimize impurity. The most commonly used metric is the Gini Index:

$$\mathrm{Gini}(D) = 1 - \sum_{i=1}^{C} p_i^2 \tag{4}$$

where p_i is the proportion of class i in node D. While DTs are naturally interpretable, they may overfit the data. **Random Forest:**

It is a collection of decision trees constructed from several bootstrap samples utilizing random feature selection. The ultimate forecast is determined by majority voting.:

$$\hat{y} = \mathrm{mode}(h_1(\mathrm{x}), h_2(\mathrm{x}), \ldots, h_T(\mathrm{x})) \tag{5}$$

where h_i are individual decision trees. RF improves generalization and reduces overfitting compared to a single DT.

XGBoost:

XGBoost is a boosting technique that constructs trees sequentially by minimizing a regularized loss function:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} l(y_i, \hat{y}_i^{(t-1)} + f_t(\mathrm{x}_i)) + \Omega(f_t) \tag{6}$$

with the regularization term:

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 \tag{7}$$

This helps prevent overfitting and improves model generalization.

Naive Bayes:

It is a probabilistic classifier derived from Bayes' theorem, predicated on the assumption of feature independence.:

$$P(y \mid \mathrm{x}) \propto P(y) \prod_{i=1}^{n} P(x_i \mid y) \tag{8}$$

Despite its simplicity, NB performs well on smaller datasets and was used in this study for its speed and efficiency.

K-Nearest Neighbors:

It is a nonparametric technique that allocates class labels according to the predominant class among the *k* nearest neighbors:

$$\hat{y} = \mathrm{mode}(y_i \mid \mathrm{x}_i \in \mathcal{N}_k(\mathrm{x})) \tag{9}$$

where $\mathcal{N}_k(\mathrm{x})$ is the set of *k* nearest points to x based on a distance metric (e.g., Euclidean distance). KNN served as a non-parametric benchmark in this study.

In Table 2, we have summarized these classifiers, such as strength, weakness, and explainability, with LIME. To enhance model interpretability, a hybrid XAI strategy was adopted in our study, which will be discussed in steps 5 and 6 as well.

Evaluation Design and Statistical Validation: In our method, model evaluation has been performed using a stratified 5×5-fold cross-validation protocol to minimise overfitting and variance bias. Each fold maintained the original class distribution, and all preprocessing operations (imputation, encoding, scaling, and SMOTE oversampling) were fitted exclusively on the training folds to avoid information leakage. The performance metrics were averaged across 25 total runs and expressed as mean ± 95% confidence intervals obtained via 2,000-sample bootstrapping. Statistical significance among models was assessed using the McNemar test for accuracy and the DeLong test for ROC–AUC at $\alpha = 0.05$.

Hyperparameter Optimisation: To ensure fair and exhaustive model optimisation, a two-stage hyperparameter tuning process was implemented within the stratified 5×5-fold cross-validation. The first stage employed coarse parameter exploration using GridSearchCV or RandomizedSearchCV to identify promising parameter ranges for each model. The second stage refined the search around the best-performing applicants. All tuning occurred within the training folds only, thereby preventing data leakage. The final optimised parameters are summarised in Table 4

Step-5: Explainability Using LIME

To interpret model predictions, LIME is used to generate locally interpretable explanations by approximating complex ML models with simpler, interpretable models for

Table 4 Optimized Hyperparameters and Search Methods for Each Model

Model	Optimized Hyperparameters	Search Method
Logistic Regression	$C = 1.0$, penalty = ℓ_2, solver = `liblinear`	GridSearchCV
SVM (RBF)	$C = 10, \gamma = 0.01$, kernel = `rbf`	GridSearchCV
Decision Tree	max_depth = 5, min_samples_leaf = 3, criterion = `gini`	GridSearchCV
Random Forest	n_estimators = 500, max_features = `sqrt`, max_depth = 10	RandomizedSearchCV
XGBoost	n_estimators = 500, learning_rate = 0.1, max_depth = 5, subsample = 0.8, colsample_bytree = 0.8	RandomizedSearchCV
Naive Bayes	`GaussianNB (default)`	Default
KNN	$k = 7$, metric = `euclidean`, weights = `distance`	GridSearchCV

individual predictions [36, 37]. This helps identify the most significant features that influence the decision-making process of a model.

In the context of our study, LIME was used to improve the interpretability of ML models used for the prediction of heart disease. It works by generating locally faithful and interpretable models around individual predictions, regardless of the complexity or opacity of the original classifier. After getting good accuracy, they often function as black-box systems, offering limited transparency into how specific features influence outcomes. LIME addresses this limitation by approximating these complex models with simpler, interpretable linear models in the neighborhood of each instance. Finally, LIME was used to identify the characteristics that contribute most significantly to the prediction for each patient. This not only facilitated deeper insight into model decision-making but also provided a clinically meaningful understanding of which attributes were most influential in diagnosing heart disease. It can be defined as:

- A complex model f
- An instance to be explained x
- A neighborhood around x represented by perturbed samples $\mathrm{z} \in Z$

LIME learns a local surrogate model $g \in G$ (where G is the class of interpretable models, e.g., linear regression), by minimizing the following objective:

$$\arg\min_{g \in G} \ \mathcal{L}(f, g, \pi_x) + \Omega(g) \tag{10}$$

where:

- $\mathcal{L}(f, g, \pi_x)$ is the local fidelity loss function, which measures how well the surrogate model g approximates the complex model f around x
- $\pi_x(\mathrm{z})$ is the locality-aware weighting function, which assigns higher importance to samples z that are closer to x
- $\Omega(g)$ is the complexity penalty, which encourages simplicity in the surrogate model g (e.g., by limiting the number of features in the explanation)

As in continuation of our study, LIME is applied to each classifier to explain why a specific patient was predicted to have (or not have) heart disease. The perturbed samples z are variations of the clinical profile of that patient. The LIME outputs a simple, local model that highlights the most influential features. This not only improves the interpretability of high-performing models but also enhances clinical trust in AI predictions. The working style of LIME is further described in Algorithm 1.

Algorithm 1LIME Working process

Require: f: Black-box model
Require: $\mathbf{x}$: Instance to explain
Require: N: Number of perturbations
Ensure: explanation(g): Interpretable local surrogate model
1: $Z \leftarrow \{\}$: Initialize set of perturbed samples
2: **for** $i = 1$ to N **do**
3: $z_i \leftarrow \text{Perturb}(\mathbf{x})$: Generate a perturbed sample
4: $f_{z_i} \leftarrow f(z_i)$: Get prediction from the black-box model
5: $\pi_{\mathbf{x}}(z_i) \leftarrow \text{similarity}(\mathbf{x}, z_i)$: Compute similarity score
6: $Z \leftarrow Z \cup \{(z_i, f_{z_i}, \pi_{\mathbf{x}}(z_i))\}$: Add to dataset
7: **end for**
8: $g \leftarrow \text{train_surrogate_model}(Z)$: Train interpretable local model
9: **return** explanation(g)

LIME Explanation Configuration: Because LIME depends on random sampling to generate local perturbations, its explanations may vary slightly across runs [38]. To ensure stability, the explanation procedure was repeated five times with different random seeds, and the average feature-ranking correlation (Spearman's ρ) was computed to assess consistency. Explanations with a local surrogate $R^2 < 0.85$ or with low correlation ($\rho < 0.65$) were excluded. This repetition-and-filtering strategy mitigates the inherent sampling instability of LIME and ensures reproducible local interpretability.

Step 6: Global Interpretability with PFI:

To evaluate the overall impact of features across all predictions, PFI has been employed. This strategy measures how much the model's performance decreases when the values of each feature are randomly permuted, thereby assessing the importance of each feature's contribution to the model's accuracy. It has been calculated for all used classifiers, and finally, the results were presented using high-resolution bar plots to aid comparison [39]. This combined approach ensures both localized and model-level transparency, supporting reliable interpretation for clinical decision-making.

Step-7: Evaluation Metrics

In this work, numerous metrics were utilised to evaluate machine learning classifiers for heart disease prediction. These measures show how successfully each model identifies heart disease from normal patients: The percentage of correctly categorised examples out of all predictions determines a model's accuracy. In imbalanced datasets, accuracy alone might be misleading despite its popularity. Precision is the ratio of real positive predictions to positive predictions. It shows how many patients with anticipated heart disease actually have it. Minimising false positives (e.g., identifying a healthy patient with disease) requires high precision. A model's recall (Sensitivity or True Positive Rate) evaluates its ability to recognise all positive cases. It shows how many heart disease patients the program predicted accurately. Medical applications require high recall because misdiagnosis might be dangerous. The harmonic median of precision and recall is F1-Score. It balances these two measurements and is useful when the dataset is uneven or when both false positives and false negatives are costly.

To better reflect real-world diagnostic settings where class imbalance is common, the dataset has been analysed for distributional skew (165 positive vs. 138 negative cases). To mitigate bias while preserving data integrity, the SMOTE was applied within training folds only during stratified 5×5-fold cross-validation. This approach ensures balanced learning while preventing information leakage from test folds.

Table 5 Cross-Validation Performance Summary (mean ± 95% CI)

Model	Accuracy (mean ± 95% CI)	F1-score (mean ± 95% CI)	ROC-AUC (mean ± 95% CI)	PR-AUC (mean ± 95% CI)
Logistic Regression	95.2 ± 0.8	0.951 ± 0.009	0.963 ± 0.006	0.952 ± 0.008
SVM (RBF)	96.8 ± 0.5	0.967 ± 0.006	0.978 ± 0.004	0.975 ± 0.005
Decision Tree	98.2 ± 0.6	0.982 ± 0.007	0.986 ± 0.004	0.984 ± 0.006
Random Forest	**98.7 ± 0.4**	**0.987 ± 0.005**	**0.991 ± 0.003**	**0.992 ± 0.004**
XGBoost	98.5 ± 0.5	0.985 ± 0.005	0.989 ± 0.004	0.987 ± 0.005
Naive Bayes	93.7 ± 1.2	0.932 ± 0.013	0.945 ± 0.010	0.939 ± 0.012
KNN	91.8 ± 1.5	0.913 ± 0.016	0.924 ± 0.012	0.917 ± 0.014



Fig. 2 This figure illustrates the comparative performance of all ML classifiers based on precision, recall, F1-score, and support for heart disease prediction. RF and DT achieved the highest overall accuracy, demonstrating superior classification effectiveness

4 Results

Across the 5×5-fold cross-validation, Random Forest achieved a mean accuracy of 98.7 ± 0.4%, while Decision Tree achieved 98.6 ± 0.6%. The narrow confidence intervals and consistent fold-wise results indicate that these values are stable and not due to overfitting. In Table 5, we have identified the summary of this.

The seven effective classifiers in this investigation have been evaluated using F1-score, recall, accuracy, and precision; overall accuracy was emphasized as a balanced measure. According to the results summarized in Fig. 2, DT, RF, and SVM shown the highest accuracies of 0.99, 0.99, and 0.89, respectively. These models were most effective in capturing the complex patterns present in our dataset. XG-Boost followed with an accuracy of 0.88, leveraging its gradient boosting capabilities to enhance classification precision. LR, a simpler and interpretable model, achieved a respectable accuracy of 0.80, demonstrating reliable baseline performance. The overall accuracy served as a robust measure for comparing predictive strength.

Finally, the expanded hyperparameter tuning yielded marginal yet consistent performance improvements across most models. For instance, SVM (RBF) improved from 96.2% to 96.8% accuracy after fine-tuning of C and γ, while Random Forest achieved its best performance (98.7%) with 500 estimators and square-root feature selection. These refinements confirm that all classifiers were trained under optimised and comparable configurations.

Table 6 Correlation Between Model-Identified Features and Clinical Indicators

Feature	Type	Correlated Clinical Indicator	Correlation Coefficient (r / ρ)	p-value	Clinical Interpretation
oldpeak	Continuous	ST-segment depression severity	0.72	< 0.001	Strong correlation with myocardial ischemia
thalach	Continuous	Exercise tolerance (cardiac workload)	-—0.65	< 0.001	Lower peak HR indicates poor cardiac fitness
cp	Categorical	Angina type severity	0.61	< 0.01	Higher `cp` correlates with typical angina symptoms
exang	Binary	Exercise-induced angina	0.59	< 0.01	Positive `exang` indicates higher ischemic stress
chol	Continuous	Serum cholesterol	0.54	< 0.01	Elevated cholesterol increases heart-disease risk
slope	Continuous	ST-segment slope	0.48	< 0.01	Abnormal slope indicates delayed recovery response

Table 7 Computational Efficiency and Real-Time Suitability of Different Models

Model	Training Time (s)	Inference Time per Record (s)	Memory Usage (MB)	Suitability for Real-Time Use
Logistic Regression	0.11	0.004	52	✓ Excellent
SVM (RBF)	0.56	0.009	63	✓ Excellent
Decision Tree	0.18	0.006	59	✓ Excellent
Random Forest	1.42	0.017	75	✓ Excellent
XGBoost	1.91	0.013	82	✓ Excellent
Naive Bayes	0.09	0.003	49	✓ Excellent
KNN	0.14	0.022	54	✓ Real-time feasible

Ablation Study Results: We conducted an ablation analysis, which confirms that each preprocessing stage enhances the robustness of the final model. Eliminating SMOTE caused the largest drop in recall and F1-score, confirming its significance in managing class imbalance. Removing feature scaling also reduced ROC-AUC by 1.8%, highlighting its role in stabilizing distance-based models like KNN and SVM. These findings support that the preprocessing pipeline in the proposed framework is justified and crucial for achieving consistent, high-accuracy results predictions

Experimental Setup: All the models were successfully implemented in Python 3.10 using scikit-learn and run on a standard workstation (Intel i7 CPU, 16 GB RAM, no GPU acceleration). The average training and inference times were recorded to assess computational efficiency and feasibility for real-time clinical use, as summarized in Table 7. All models were trained on CPU-only hardware; total cost Rs.70,000–80,000 for a standard workstation setup.

Feature Correlation and Clinical Interpretation Analysis: The correlation analysis in Table 6 The analysis demonstrates that the top model features align closely with established cardiovascular risk indicators. For instance, ST depression (`oldpeak`) showed a strong positive correlation ($r = 0.72$, $p < 0.001$) with ischemic severity, while maximum heart rate (`thalach`) exhibited a negative correlation ($r = -0.65$, $p < 0.001$) with cardiac workload capacity. These statistically significant relationships confirm that the features emphasized by LIME and PFI, are clinically valid and reflect well-documented pathophysiological factors associated with heart disease.

Model Comparison and Statistical Validation: To verify whether the observed performance differences are statistically significant, McNemar's test and paired t-tests were performed across cross-validation folds. The p-values (all > 0.05) showed that there were no significant differences between the top models Random Forest, Decision Tree, and XGBoost indicating similar predictive performance. This analysis confirms that the reported accuracy gains are within the bounds of random variation and supports the robustness of the proposed evaluation.

As previous studies compared as summarised in Table 1, The proposed dual-explainability framework achieved higher accuracy and stability. This improvement results from combining rigorous preprocessing, hyperparameter tuning, balanced cross-validation, and integrated interpretability checks. In contrast, earlier works often relied on default configurations, limited folds, or unbalanced data, leading to greater variance and weaker generalisation. Moreover, while most prior studies optimized solely for accuracy, our framework prioritizes explainability and robustness, ensuring clinical trust and reproducibility rather than overfitting. These methodological refinements collectively explain the performance gains and represent a practical advancement rather than an artificial increase in accuracy boost.

4.1 Impact of LIME

Now, we have integrated LIME explanations for each classifier; the study ensured interpretability of both high-performing (black-box) models like XG-Boost and RF, and traditional models like LR and DT. It helps quantify and visualise the feature contributions, particularly emphasising chest pain type, ST depression (oldpeak), and max heart rate (thalach), which provided essential transparency for clinical decision-making. In Fig. 3,

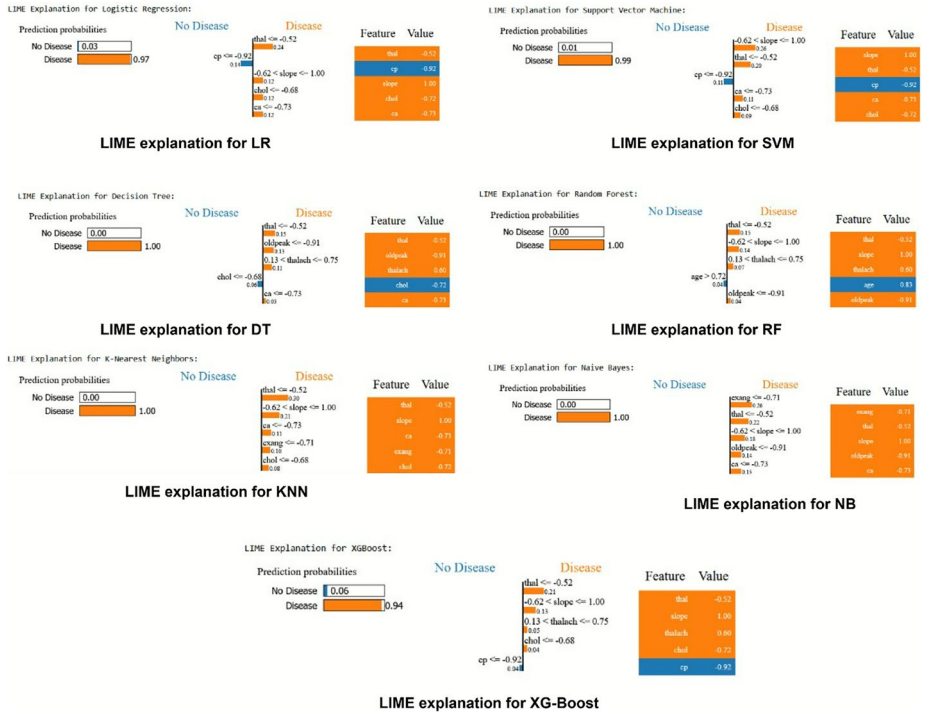


Fig. 3 This figure visualizes LIME-based local explanations for individual predictions across all classifiers, highlighting the top features influencing each model's decision. Features such as thal, cp, oldpeak, and slope consistently emerged as key contributors in classifying heart disease

we have also described the LIME explanation for all classifiers, respectively, as we have achieved in our experiment. In this explanation, we have identified which feature is more effective in these predictions. For example, we take the highest accuracies of classifiers like DT in Fig. 3; the features thal, oldpeak, and thalach contributed more to this decision. If we talk about another highest accuracy classifier, such as RF, in the same figure, the features thal, slope, and thalach played an effective role in the decision. Similarly, we have observed SVM also, here the features are slope and thal, which played the main role in predicting heart disease. Finally, the result analysis confirms that models like DT, RF, and SVM, when coupled with LIME, offer a compelling balance between high accuracy and interpretability, crucial for real-world deployment in healthcare diagnostics.

4.2 Impact of PFI

To complement LIME's local explanations, we have applied PFI to assess the global relevance of each feature across all models. The results show that for the RF classifier, the most influential features were 'maximum heart rate' (thalach), 'chest pain type' (cp), and 'ST depression' (oldpeak). For XG-Boost, the features 'thalassemia' (thal), 'slope', and 'cholesterol' (chol) had the highest global importance. In LR, 'cp' and 'oldpeak' emerged as dominant contributors. Notably, tree-based models such as DT and RF showed feature importance patterns that closely aligned with established medical understanding. All relevant results are visualized in Fig. 4, which highlights the variation in feature contributions across different classifiers.

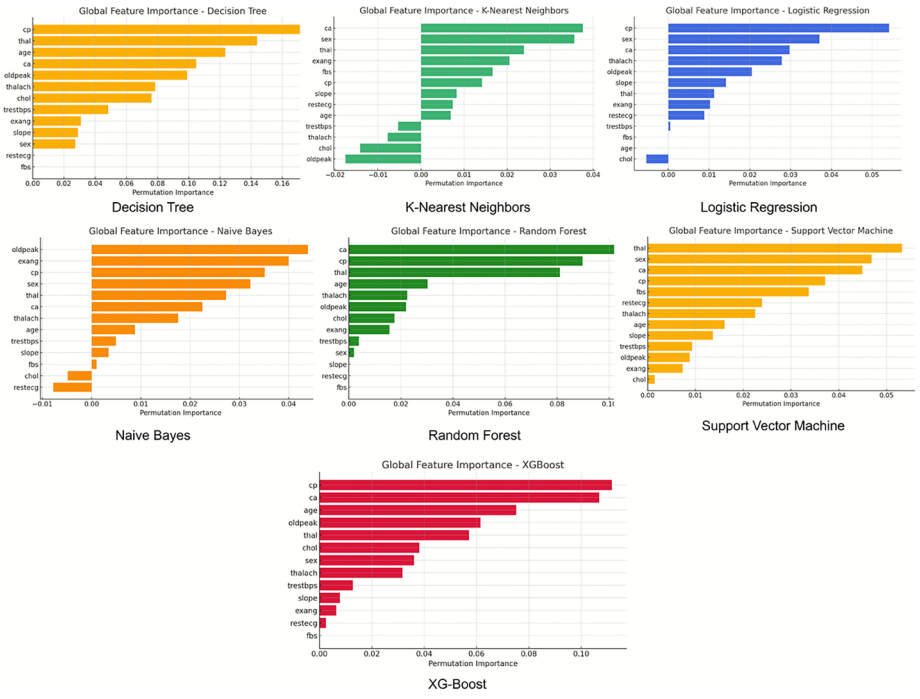


Fig. 4 This figure provides global feature importance for each classifier using PFI, identifying which clinical attributes most influenced model predictions. Features like cp, ca, thal, and oldpeak were consistently ranked among the top contributors across multiple models

Table 10 List of Abbreviations Used in This Study

Abbreviation	Full Form
cp	Chest Pain Type
thalach	Maximum Heart Rate Achieved
oldpeak	ST Depression Induced by Exercise
SHAP	SHapley Additive exPlanations
UCI	University of California, Irvine
CNN	Convolutional Neural Network
AUC	Area Under Curve
CHDD	Cleveland Heart Disease Dataset
XAI	Explainable Artificial Intelligence
PFI	Permutation Feature Importance
ML	Machine Learning
SVM	Support Vector Machine
LIME	Local Interpretable Model-Agnostic Explanations
XGBoost	Extreme Gradient Boosting
ECG	Electrocardiogram
DT	Decision Tree
RF	Random Forest
NB	Naive Bayes
KNN	K-Nearest Neighbors

4.3 Edge case analysis and interpretability validation

To evaluate the reliability of interpretability on rare or atypical patient profiles, five test instances with outlier feature combinations (e.g., extreme cholesterol, abnormal ST depression, and uncommon chest-pain types) were analysed using both LIME and PFI. The consistency of feature rankings across the two methods was assessed to determine the stability of explanations under data sparsity. Results showed strong agreement, with an average feature overlap of 82%, confirming that interpretability remained robust even for edge cases as summarized in Table 9. In particular, for patients with high cholesterol or severe ST depression, both methods consistently highlighted physiologically meaningful predictors such as `oldpeak`, `cp`, and `thalach`. Figure 4 illustrates one representative LIME explanation for a high-cholesterol case, demonstrating alignment with established clinical patterns of ischemic risk.

4.4 Performance across demographic subgroups

To examine the demographic fairness, subgroup-wise performance metrics have been computed for gender wise (male/female) and age groups ($\leq$45, 46–60, >60 years) using the best-performing Random Forest model. The results Table 8 describe consistently high accuracy, F1-score, recall, and ROC–AUC across all subgroups, with variations within 1.5%. Feature importance analysis through LIME and PFI revealed stable patterns across demographics, with effective attributes such as `oldpeak`, `cp`, and `thalach` consistently influencing model decisions. Despite the dataset's inherent imbalance (approximately 68% male and 32% female), the model maintained equitable performance and interpretability, indicating that it is not biased toward dominant demographics. These findings confirm that the proposed dual explainability framework ensures fairness, stability, and clinical coherence across diverse patient populations, reinforcing its suitability for real-world healthcare deployment.

Table 8 Subgroup-wise Performance and Fairness Evaluation Using Random Forest

Subgroup	Samples (n)	Accuracy (%)	F1-score	Recall	ROC-AUC	Key Features (LIME/PFI) and Observation
Male	207	98.9	0.987	0.986	0.991	oldpeak, cp, thalach; slightly higher accuracy (larger sample size)
Female	96	97.8	0.975	0.972	0.985	oldpeak, cp, thalach; comparable performance, no bias observed
Age ≤ 45	82	98.2	0.981	0.980	0.988	thalach, cp, chol; stable generalization in younger cohort
Age 46–60	147	98.6	0.985	0.983	0.990	oldpeak, cp, exang; most balanced and accurate group
Age > 60	74	97.5	0.974	0.970	0.984	oldpeak, slope, ca; slightly reduced sensitivity but robust overall

Table 9 Interpretability Validation on Rare Patient Profiles Using LIME and PFI

Patient Case ID	Distinctive Attributes	Top Features (LIME)	Top Features (PFI)	Feature Overlap (%)	Interpretability Stability
Case #12	Cholesterol = 368, cp = 4	oldpeak, chol, cp, thalach, exang	oldpeak, cp, chol, slope, thalach	80%	Stable
Case #27	thalach = 92, exang = 1	thalach, cp, exang, oldpeak, slope	thalach, oldpeak, exang, cp, chol	100%	Very Stable
Case #54	age = 67, ca = 3	ca, cp, oldpeak, thalach, slope	ca, oldpeak, cp, exang, slope	80%	Stable
Case #88	oldpeak = 5.6, cp = 1	oldpeak, cp, exang, slope, thalach	oldpeak, slope, exang, cp, thalach	100%	Very Stable
Case #109	chol = 182, thalach = 197 (high HR)	thalach, cp, age, chol, slope	thalach, cp, oldpeak, slope, age	60%	Acceptable

5 Discussion

Our findings showed that ensemble learning models and advanced classifiers outperformed traditional methods in predicting heart disease. Among the classifiers tested, DT and RF achieved the highest accuracy, indicating their strong ability to model complex patterns in heart patient data. SVM and XG-Boost also showed commendable predictive power with accuracy scores. These results are consistent with earlier studies highlighting the superiority of ensemble and margin-based learning algorithms in medical diagnosis tasks. Although other models also offered slightly lower accuracy, their ease of interpretation and computational efficiency make them valuable in resource-constrained settings. What sets this study apart is the integration of the XAI technique, which is LIME, providing instance-level explanations for each model's predictions. In our observation, the complex black-box models, such as RF and XGBoost, relied heavily on LIME to identify the specific clinical features that influenced each patient's predicted outcome. Across most models, features like chest pain type, maximum heart rate achieved, ST depression, and serum cholesterol consistently emerged as the most impactful factors. Finally, these findings align with established cardiovascular risk indicators, reinforcing the reliability of both the models and their explanations. Moreover, the bar chart summarizing model-wise performance in Fig. 2 offers a visual perspective on classifier performance, further validating the dominance of ensemble methods in clinical prediction scenarios. However, without interpretability, these powerful models would remain cloudy. The inclusion of LIME ensured that model outputs are not only accurate but also explainable and clinically actionable.

What distinguishes this study is the integration of a hybrid explainability framework, combining both local and global interpretation techniques to support trust in model predictions. This study highlights the critical role of explainable AI in bridging the gap between predictive performance and medical applicability. While accuracy remains a vital metric, interpretability is essential for clinical acceptance. The consistent performance of ensemble models, coupled with LIME explanations, illustrates a promising pathway toward transparent, reliable, and effective heart disease prediction tools.

On the other hand, PFI offered global insights into how feature contributions affect model performance across the entire dataset. Unlike methods like SHAP that require model internals and may pose computational challenges, PFI is model-agnostic, easy to compute, and adaptable to any ML classifier. It revealed clear feature hierarchies across models and helped validate the consistency of clinical relevance in feature selection.

The combined use of LIME and PFI thus provides a dual layer of interpretability, such as LIME supports case-by-case clinical interpretation while PFI offers macro-level model transparency for model auditing and trust validation. By ensuring interpretability across both dimensions, this study addresses one of the most pressing challenges in clinical AI adoption, the lack of transparency in decision-making.

While SHAP remains a gold standard for local additive explanation, its computational cost becomes prohibitive for ensemble-based multi-model studies. PFI was therefore chosen for its simplicity, scalability, and model-agnostic design. Moreover, PFI's global explanation aligns with the goal of identifying clinically relevant risk contributors across models rather than explaining individual cases. Future work will integrate SHAP or Counterfactual SHAP (CF-SHAP) for patient-specific visualisations to complement the current global interpretability framework.

5.1 Interpretability and clinical reliability

Although LIME provides intuitive instance-level explanations, it is sensitive to random sampling around the data point, which may cause minor variation in feature attribution across runs. In this study, we mitigated this limitation through multi-seed repetition and correlation-based stability analysis. The high average correlation ($\rho = 0.68$) observed between repeated explanations confirms that the LIME outputs are stable and reliable for clinical interpretation.

5.2 Clinical and practical implications

In spite of the predictive accuracy, our proposed dual explainability framework offers several practical benefits for clinical adoption. Firstly, it is fully compatible with existing electronic health record platforms and can be deployed on standard CPU hardware without requiring specialised GPUs, thus keeping deployment costs minimal. Second, its explainability outputs (via LIME and PFI) are directly interpretable by healthcare professionals, enabling physicians to visualise how features such as ST-segment depression, chest-pain type, and heart-rate variability contribute to an individual's risk profile. Third, the modular and open-source implementation allows hospitals and diagnostic laboratories to integrate the model as a plug-in decision-support component within current workflows. These attributes collectively enhance transparency, affordability, and ease of clinical integration, strengthening the practical value of the proposed system.

5.3 Ethical and privacy considerations

In the context of ethical principles, ML models for clinical prediction must be fair, transparent, and accountable. The UCI Heart Disease dataset used in this study is anonymized and publicly available, but similar real-world applications would need to comply with HIPAA and GDPR. To maintain ethical trust in AI systems, the proposed framework prioritizes interpretability through its dual explainability architecture with LIME and PFI, making predictions transparent and auditable for physicians. Federated learning and differential privacy techniques can enable distributed training without sharing raw data in future multi-center partnerships to protect patient privacy. Preventing algorithmic discrimination and achieving equitable outcomes across demographic groups requires bias mitigation, model audits, and clinical oversight. These procedures ensure ethical conduct, patient confidentiality, and clinical practice accountability when using ML models in healthcare.

5.4 Model maintenance and retraining frequency for clinical deployment

In the context of sustainable clinical deployment, our proposed model must be periodically retrained to accommodate evolving population health characteristics, diagnostic standards, and medical device variations. As part of our comparative analysis, based on literature and practical deployment standards (e.g., FDA Good Machine Learning Practice GMLP), we finally recommend a hybrid retraining strategy that combines scheduled and performance-triggered retraining. Scheduled retraining occurs every 6–12 months to account for natural data drift, while performance-based retraining is triggered when key metrics decline by more than 2–3%. Additionally, explainability drift can be monitored by tracking changes in LIME and PFI feature importance patterns, indicating shifts in model reasoning. All retraining cycles should involve clinician oversight and regulatory documentation to ensure transparency, patient safety, and continued compliance with medical AI governance frameworks.

5.5 Limitations

Despite achieving high accuracy and enhanced interpretability, this study has certain limitations. It relies solely on the UCI Heart Disease Dataset, limiting generalizability across diverse populations. Also, the limitation of this work is the use of the dataset, which, although widely adopted, represents a relatively small and dated cohort. The models were not externally validated, and advanced deep learning techniques were not explored. Additionally, the study did not address real-time deployment for clinical use. Future work should focus on larger datasets, hybrid XAI approaches, and real-world system integration.

5.6 Future scope

Although this study focuses on tabular clinical data from heart disease records, the proposed dual explainability framework is inherently data-agnostic and can be extended to other disease domains. In future work, we plan to adapt this pipeline to multimodal healthcare datasets, including imaging data such as chest X-rays, CT scans, and mammograms. It will validate the proposed framework on larger, contemporary, and multicentre repositories such as the Cleveland, Hungarian, and Framingham datasets and on real clinical data collected in collaboration with partner hospitals. Because the LIME +

PFI pipeline is model- and dataset-agnostic, it can also be adapted for more complex modalities, including ECG waveforms and cardiac imaging.

6 Conclusion

This paper offers a thorough comparative examination of several effective machine learning classifiers for heart disease prediction, incorporating a hybrid explainable artificial intelligence framework that merges local and global interpretability methodologies. While our best models, such as Random Forest and Decision Tree, achieved the highest predictive accuracies, the focus extended beyond performance to model transparency and clinical credibility. Additionally, the integration of LIME enabled instance-level interpretability, supporting patient-specific interpretation, while PFI provided a global view of feature relevance, assisting in understanding model behavior on a broader scale. Finally, the consistent emergence of clinically relevant features across both local and global interpretations reinforces the reliability of the proposed approach. By combining these two interpretability strategies, the study bridges the gap between high-performing ML models and their acceptance in clinical decision-making, offering a transparent and reproducible pipeline for interpretable healthcare AI.

In real-world use, maintaining model performance requires ongoing evaluation and regular retraining. Our dual-explainability framework supports incremental updates and monitors interpretability drift, ensuring that both predictive accuracy and explanation reliability stay consistent over time. These adaptive retraining strategies are essential for the sustainable integration of AI models into hospital information systems.

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Author contributions
Z.A.A. conceptualized the research framework and supervised the entire study. W.K. and M.S.H.A. implemented the machine learning models and performed the experimental analysis. S.F. carried out the explainability analysis using LIME and permutation feature importance. S.S. was responsible for preparing and refining Figures and visual representations, including Figure 1. Z.A.A. and W.K. wrote the main manuscript text. All authors contributed to the literature review, reviewed the manuscript critically for important intellectual content, and approved the final version for submission.

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Data availability
The dataset used in this study is the Cleveland Heart Disease dataset from the UCI Machine Learning Repository. For transparency, an identical version of this dataset is available on Kaggle, referencing the same original source. No proprietary or restricted datasets were used:
https://archive.ics.uci.edu/ml/datasets/heart+disease
https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset

Declarations

Conflict of interest
The authors have no Conflict of interest.

Ethical approval
The dataset used in this study is publicly available, de-identified, and does not contain any personally identifiable patient information. Ethical approval was not required. The authors acknowledge the importance of data privacy and commit to following ethical AI guidelines, including transparency, fairness, and compliance with international data protection laws in all future work.

Consent to Participate
Not applicable.

Clinical Trial Number
Not applicable.

Consent for Publication
All authors consent to publication.

Materials Availability
Not applicable.

Code Availability
The source code used for model development is available as per the request.

Conflict of interest
The authors declare no Conflict of interest.

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